

Open-Loop L2 Is Blind: Planning Metrics Hide Quantization Damage in Autonomous-Driving Vision–Language–Action Models

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Abstract—Post-training quantization is routinely reported “lossless” for autonomous-driving vision-language-action (VLA) models using the standard open-loop nuScenes L2 metric. We show this conclusion is an artifact of the metric. Using OpenDriveVLA-0.5B (a Qwen2.5 planner on UniAD perception), we first reproduce the lossless result—FP16, INT8, and INT4 weight quantization of the language backbone all yield ≈ 0.33 m L2. We then run a single controlled ablation: zeroing the ego-state and history text handed to the planner inflates L2 from 0.33 to 6.38 m ($19\times$). Open-loop planning on nuScenes is therefore $\approx$ ego-motion extrapolation; the perception/planning pathway—and any quantization damage to it—contributes little to the reported number, which is why model size and bit-width barely move it. The practical implication is a warning: do not certify a quantized driving policy on open-loop L2. We outline why closed-loop testing is necessary to expose quantization-induced failures that open-loop metrics structurally cannot see.

Index Terms—vision-language-action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

Compressing driving VLA policies for on-board inference is attractive, and the literature generally reports that 4-bit weight quantization is essentially free on open-loop nuScenes planning. We argue the standard metric cannot support that claim. We make three points, each backed by a measurement on OpenDriveVLA-0.5B: (i) weight quantization down to INT4 is indeed lossless on open-loop L2; (ii) open-loop L2 is dominated by ego-motion extrapolation, so it is structurally insensitive to the perception/planning computation; (iii) therefore the “lossless” conclusion says nothing about deployment behavior.

II. Setup

OpenDriveVLA-0.5B couples a Qwen2.5-0.5B language/action head (hidden size 896, 24 decoder layers) to UniAD stage-1 track+map perception used as a “vision tower.” We evaluate on the 162 nuScenes-mini $\cap$

validation samples under the ST-P3 and UniAD L2 protocols. Weight-only post-training quantization (per-output-channel symmetric round-to-nearest, dequantized to the compute dtype) is applied only to the 168 Linear layers of the Qwen backbone (357.8 M parameters); perception, the scene/track/map projectors, embeddings and the output head remain in floating point, so the measured effect is purely the language head’s Wb loss.

III. Results

A. INT4 is “lossless” on open-loop L2

Down to INT4 the planner is indistinguishable from FP16. The degradation at INT3/INT2 is a generation-coherence failure (malformed trajectory text), not graded planning loss.

B. Why the metric cannot see quantization: ego-ablation

Removing the handed ego kinematics collapses the plan by $19\times$, while the outputs remain well-formed. The model is therefore extrapolating the ego trajectory it is given; perception and the language head contribute little to open-loop L2. This single number explains the otherwise puzzling observations that (a) 0.5B/3B/7B variants report the same L2 and (b) INT8/INT4 are free: there is little perception-grounded reasoning in this metric to lose.

a) On the deployment target.: We replayed the captured planner inputs on a real Jetson AGX Orin 64 GB (JetPack 6, torch 2.8) and re-scored open-loop L2 (40-sample mini subset, ST-P3). The Orin reproduces the A100 to ≈ 1 cm mean (FP16), and INT4 remains “lossless” on-device (FP16 0.87 m vs group-INT4 0.90 m at 3 s)—even though the per-sample W4 plans already differ from FP16 by 14–24 cm. The blindness travels to the edge hardware: it is the metric, not the machine.

IV. Discussion

Open-loop L2 measures how well a model copies a velocity vector forward, not how well it perceives or plans. Quantization that subtly perturbs perception or

TABLE I

Open-loop L2 (m, ST-P3 cumulative protocol; lower is better), 162 miniVal samples. Weight-only quantization of the Qwen backbone.

Precision	L2 @1/2/3s avg	well-formed output	note
FP16 (baseline)	0.33	100%	matches reported 0.5B result
INT8	0.33	100%	lossless
INT4	0.33	100%	lossless
INT3	1.91	92%	text generation begins to break
INT2	6.45	0%	dead (all malformed)

TABLE II

Ego-ablation: zeroing the ego-state and 2s history text in the prompt. Same model, same perception.

Condition	L2 (m)	relative
Full input (FP16)	0.33	1×
Ego-state + history zeroed	6.38	19×

planning—or that destabilizes generation under distribution shift—is invisible to it. Certifying a quantized driving policy on open-loop L2 thus provides false assurance. The necessary test is closed-loop: only there is the logged ego trajectory unavailable to coast on, so perception/planning, and quantization damage to them, can manifest. A companion study shows exactly such a failure (per-channel INT4 collapsing generation closed-loop where group-wise INT4 does not), invisible to the L2 metric here.

V. Limitations

The ego-ablation and quantization sweep are on a single model (OpenDriveVLA-0.5B) and the nuScenes-mini split. The claim is about the metric’s structural insensitivity, supported by the 19× ablation; it does not assert a specific quantization technique is unsafe (that is the subject of the closed-loop companion work).

VI. Conclusion

4-bit weight quantization of a driving-VLA language head is lossless on open-loop nuScenes L2—but the metric is $\approx$ ego-extrapolation (19× ego-ablation), so “lossless” is not evidence of safety. Quantized driving policies must be evaluated closed-loop.

References

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